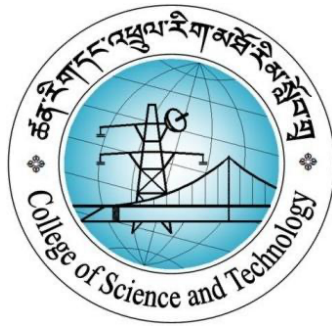


SWE 101 Report



Second Year Semester I

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1. Introduction

This project involves the development of a mobile application called Campus Companion using React Native with Expo. The main goal of this app is to help new students easily access important information about the college, such as contacts, class schedules, and campus resources.

The application is designed to be simple, easy to use, and responsive so that it works properly on different screen sizes. It also follows a clean structure so it can be improved in future assignments.

2. Objective of the Project

The objectives of this project are:

- To understand the basic structure of a React Native application
- To use core components such as View, Text, ScrollView, and FlatList
- To implement navigation between multiple screens
- To design a user-friendly interface using proper styling
- To make the application responsive on different devices

3. Features of the Application

The Campus Companion app includes the following features:

- Home Screen
Displays the application title and a welcome message.
- Contacts Screen
Shows a list of important contacts such as IT Helpdesk and Student Services using a FlatList.
- Contact Details Screen
When a contact is selected, detailed information like name and phone number is displayed.
- Schedule Screen
Displays a simple weekly class timetable using a scrollable view.
- Map Screen
Provides a button that opens the campus location in Google Maps.

4. Application Structure

The project is organized into different folders to maintain clean and readable code:

- screens/ – contains all screen components
- components/ – contains reusable components
- navigation/ – handles navigation setup
- data/ – stores static data such as contacts
- types/ – contains TypeScript definitions

This structure makes the application easier to understand and modify.

5. Implementation Details

The application is built using multiple screens connected through navigation:

- A Bottom Tab Navigator is used to switch between main screens such as Home, Contacts, Schedule, and Map.
- A Stack Navigator is used to navigate to the Contact Details screen.
- The FlatList component is used to efficiently display a list of contacts.
- Data is passed between screens using navigation parameters when a contact is selected.

For styling:

- The layout is created using Flexbox
- Styles are applied using StyleSheet
- A dynamic style is implemented to highlight a selected contact item

For responsiveness:

- Flexible layouts are used so the app adjusts to different screen sizes
- ScrollView and FlatList ensure content does not overflow on smaller devices

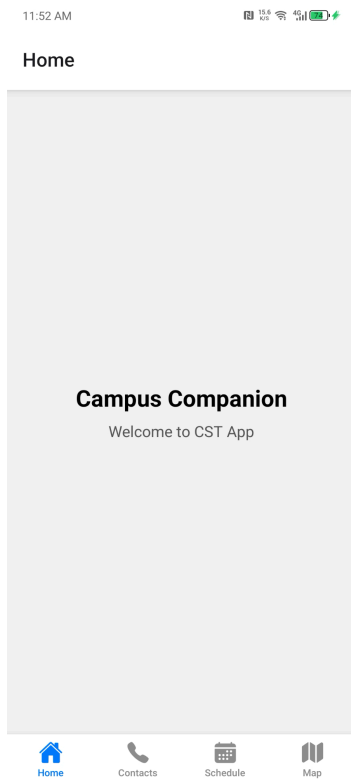
6. User Interface Design

The user interface is designed to be simple and clear:

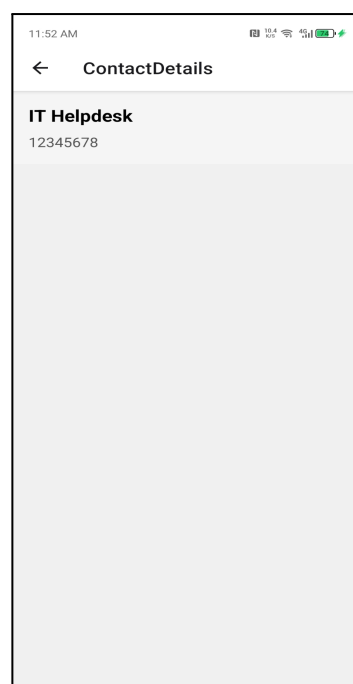
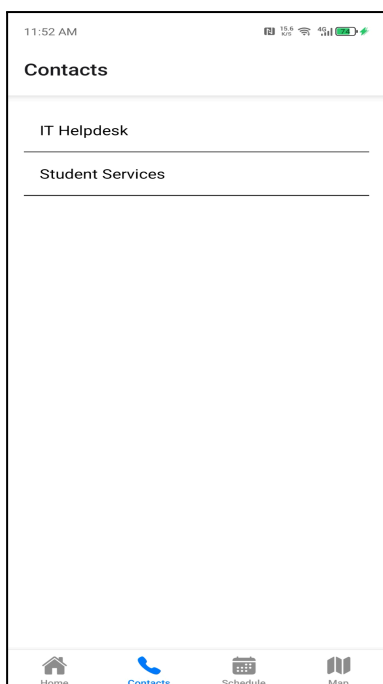
- The layout is clean and easy to navigate
- Text is readable and properly spaced
- Navigation between screens is smooth
- The app works well on both small and large devices

7. Screenshots

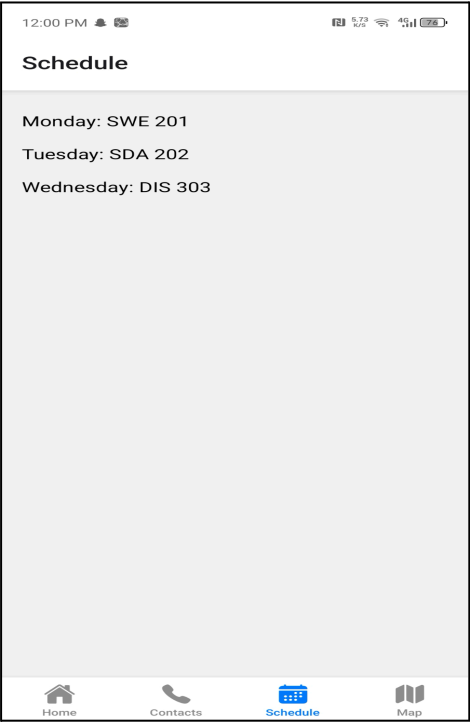
- **Home Screen**



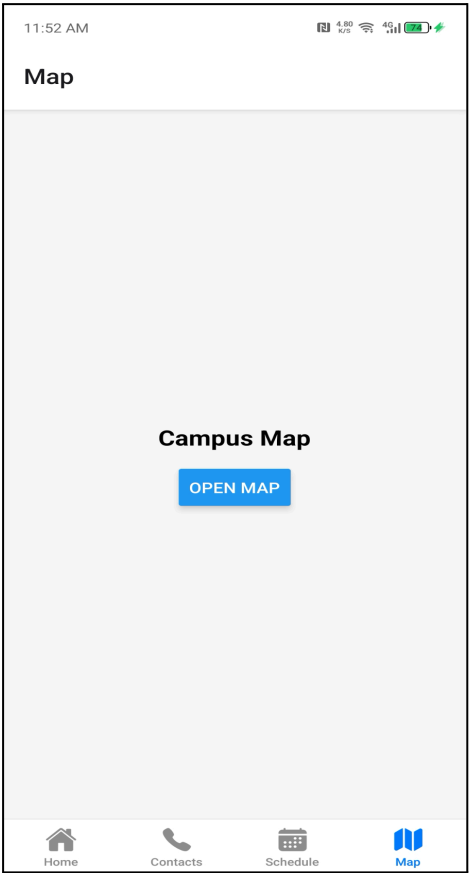
- **Contacts Screen & Contact Details Screen**



- **Schedule Screen**



- **Map Screen**



8. Challenges Faced

During the development of this project, a few challenges were encountered:

- Understanding how to set up navigation using React Navigation
- Managing TypeScript types for navigation and components
- Organizing files into proper folders

These challenges were solved through practice and testing.

9. Conclusion

In conclusion, the Campus Companion app successfully meets the requirements of the assignment. It demonstrates the use of React Native components, navigation, styling, and responsive design.

This project helped in gaining a better understanding of mobile app development and provides a strong foundation for building more advanced applications in the future.